

Machine Learning-Based Prediction of Toughness in Fused Filament Fabrication: Leveraging In-Process Annealing With Enhanced Printheads

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Abstract

Fused Filament Fabrication (FFF) is widely adopted for its affordability and design flexibility, but printed Acrylonitrile Butadiene Styrene (ABS) parts often exhibit poor Z-direction strength due to weak interlayer adhesion and residual stresses. To address this, an enhanced printhead enabling in situ annealing was employed to promote improved bonding through controlled thermal management. This study leverages machine learning (ML) to predict mechanical toughness from thermal features captured by infrared thermography. Forty-four thermal features were analyzed using correlation and Principal Component Analysis, retaining 99.8% variance. Multiple classifiers, including k-Nearest Neighbors, Decision Trees, Naïve Bayes, Support Vector Machines, Artificial Neural Networks, XGBoost, and CatBoost, were evaluated. Artificial Neural Networks achieved the highest balanced accuracy (86.7%) and Area Under the Curve (90.7%) for toughness classification. Regression models however, offered limited continuous predictions. The integration of thermal monitoring with ML shows strong potential for non-destructive, real-time quality assurance in enhancing thermoplastic additive manufacturing reliability.

1. Introduction

Among different Additive Manufacturing (AM) techniques FFF has witnessed rapid adoption due to its cost effectiveness, simplicity, accessibility and versatility [1], [2], [3], [4]. These attributes have positioned thermoplastic-based FFF as a prominent method for fabricating functional components across sectors such as aerospace [5], [6], biomedical engineering [7], [8], marine [9], [10] and automotive manufacturing [11]. However, one of the limitations of FFF is its anisotropic mechanical performance, with notably lower strength in the Z-direction. This weakness stems from insufficient bonding between adjacent layers during the printing process [12], [13].

This problem is particularly acute in Acrylonitrile Butadiene Styrene (ABS), a common amorphous thermoplastic used in FFF, which exhibits a relatively high and broad glass transition range due to its heterogeneous phase composition (~100-110 °C). During deposition, rapid cooling and thermal creep inhibit adequate polymer chain entanglement across layers, inducing residual stresses that manifest as voids, delamination, and poor toughness in the build (Z) direction [14], [15]. Dynamic mechanical analysis (DMA) studies have corroborated these effects, showing a markedly lower storage modulus along the Z-axis compared to the planar directions [15]. To counteract these shortcomings, researchers have explored both material-level modifications [16] and process optimization strategies involving raster angle, build orientation, and extrusion temperature [17]. While such approaches yield partial improvements, they are typically constrained by limited adaptability and real-time response during manufacturing. In response, recent hardware innovations have introduced enhanced printhead systems capable of in-situ annealing, delivering localized thermal energy during deposition to improve interlayer bonding and improve residual stress without the need for further post-processing [18], [19]. Simultaneously,

machine learning (ML) is increasingly being used for correlating process parameters, such as temperature and print speed, with part quality, thereby enabling data-driven quality prediction frameworks [20], [21], [22]. However, most previous studies analyzed the parts only after printing is complete and rarely combine in-situ annealing with real-time prediction of mechanical toughness especially for thermoplastic FFF materials like ABS.

This study aims to address the identified gap by integrating localized thermal control through a patent-pending custom-designed printhead with machine learning models trained on thermal process data to predict mechanical toughness in ABS specimens. A total of 44 engineered features were derived from layer-wise printing and cooling profiles, capturing critical thermal characteristics such as interlayer gradients, cooling rates, and transient behavior. These features were reduced using Principal Component Analysis (PCA) to retain essential variance while minimizing dimensionality. The resulting components were used to train multiple supervised classifiers, enabling high-accuracy prediction of part toughness. The proposed framework demonstrates the potential of combining in-situ thermal control with data-driven modeling for intelligent, quality-aware FFF processing.

2. Related Work

Considering FFF's potential various studies have explored process optimization strategies such as raster angle tuning, build orientation changes, extrusion temperature control, and print speed adjustments to improve strength and reduce anisotropy [23], [24], [25]. While these approaches have led to incremental improvements, they remain constrained by fixed parameters and offer limited adaptability during printing.

To overcome limitations in interlayer adhesion, several in-situ annealing techniques have been explored. One study employed a focused laser beam to selectively preheat the interface between deposited tracks, raising the local temperature above the polymer's glass transition temperature (T_g). This approach enhanced chain reptation and molecular diffusion, thereby improving weld strength [26]. Another study used directed heated air to convectively raise the temperature of the newly deposited layer or the nozzle-substrate interface, maintaining temperatures above T_g to facilitate stronger fusion [27]. Additionally, infrared (IR) radiation has been used to locally heat the build surface via radiative absorption while minimizing thermal exposure to the entire part [27]. A recently developed printhead incorporating directional heating via a heated metal plate during deposition has demonstrated significant improvements in both tensile strength and fracture toughness for PLA parts [18]. While these localized heating strategies offer precise thermal control, they are typically not integrated with adaptive data-driven decision systems.

In parallel with hardware innovation, machine learning (ML) has emerged as a promising tool for modeling and predicting mechanical properties in FFF. Compared to conventional simulation methods, trained ML models offer rapid output, making them well-suited for real-time control applications [28]. This has led to increased interest in integrating thermal and visual signals into predictive ML frameworks. One study trained a Random Forest Regressor using in-situ infrared thermal data to predict tensile strength, revealing that cooling rates influenced by infill orientation play a key role in interlayer bonding [29]. A review further emphasized the importance of cooling-stage temperature profiles, highlighting their impact on strength, stiffness, and dimensional stability. The work noted the scarcity of layer-specific thermal studies and advocated for infrared thermography as a non-contact tool to monitor T_g and detect defects [30].

Although direct in-situ annealing is rarely the primary focus in ML-based FFF studies, thermal parameters are often treated as critical input features. One study applied polynomial regression to predict the energy absorption of PETG-based auxetic structures using printing temperature, print speed, and layer height, achieving an R^2 of 92.46% [31]. While not targeting toughness directly, the work noted PETG's suitability due to its high toughness. Another investigation used ML models to explore the effects of post-process annealing and nozzle diameter on the tensile and flexural strength of PETG and PETG-CF composites [32]. In another work in-situ vibration and current sensing were coupled with a GA-BPNN model to predict compressive strength, illustrating how real-time process dynamics influence part quality [33]. In a related domain, fracture toughness of pultruded composites was predicted using surrogate mechanical properties in a data-driven ML framework, demonstrating high predictive accuracy and the feasibility of toughness estimation [34].

Collectively, these works illustrate the expanding role of ML in mechanical performance prediction for FFF. However, the integration of engineered thermal features, active in-situ annealing, and ML-based toughness prediction remains largely unexplored. The present study addresses this gap by developing a data-driven classification framework that fuses localized thermal control with supervised ML models to predict mechanical toughness in real time.

3. Experimental Setup and Data Collection

This section outlines the complete experimental and data processing workflow employed in this study, consisting of specimen fabrication, thermal data acquisition, mechanical testing, and feature generation for machine learning analysis. Subsection 3.1 introduces the material system and describes the hardware configuration, including the use of a custom-developed enhanced printhead designed to enable directional in-situ annealing during FFF. Subsection 3.2 details the thermal imaging setup and procedures, where an infrared (IR) camera was used to capture real-time interface temperature profiles between deposited layers under varying print conditions. In subsection 3.3, the destructive testing methodology is explained, including tensile testing performed according to ASTM standards to compute the toughness of each specimen. Lastly, subsection 3.4 presents the thermal feature engineering process, describing how 44 features were systematically extracted from both the printing and cooling phases to characterize interlayer thermal behavior relevant to bonding quality and mechanical performance.

3.1. Material and Printing System

Two Creality Ender 3 Pro printers were utilized, one with a stock printhead featuring a standard 0.5 mm nozzle and the other with enhanced printhead assembly to enable in-process thermal annealing. The enhanced printhead assembly illustration is shown in Figure 1 with three consecutive layers marked which are focused in this study. To maintain constant temperature during the printing both printers were kept inside enclosures. Specimens were fabricated using 1.75 mm diameter black-colored ABS filament (Hatchbox 3D, Pomona, CA, USA). Due to the amorphous nature of ABS exhibit a broad range of T_g instead of single value [35]. Thus, differential scanning calorimetry (DSC) was used to quantify T_g in accordance with ASTM E1356 guidelines. Using the onset method, T_g was determined to be 110.21 °C.

The print parameters optimized for print quality and mechanical integrity are listed in Table 1. Specimens were printed vertically (aligned in the z-direction) to explicitly test the weakest interlaminar bonds. Three evenly spaced dog bones were printed simultaneously during every

print. Different distances and print speed were selected to vary the print time and cool time of dog bones and their effect on inter-layer bond strength.

Table 1: Print parameters

Parameter	Value	Parameter	Value
Nozzle Temperature	240°C	Bed Temperature	100°C
Infill Raster Orientation	0°	Extrusion width	0.55 mm
Extrusion Multiplier	1.1	Layer Height	0.25 mm
Retraction	6.5 mm	Outline overlap	80%
Outline	1	Infill	100%
Print Speed	1200mm/min, 1800mm/min, 2400mm/min	Dogbone Spacing	7.5mm, 12.5mm, 25mm, 50mm, 75mm

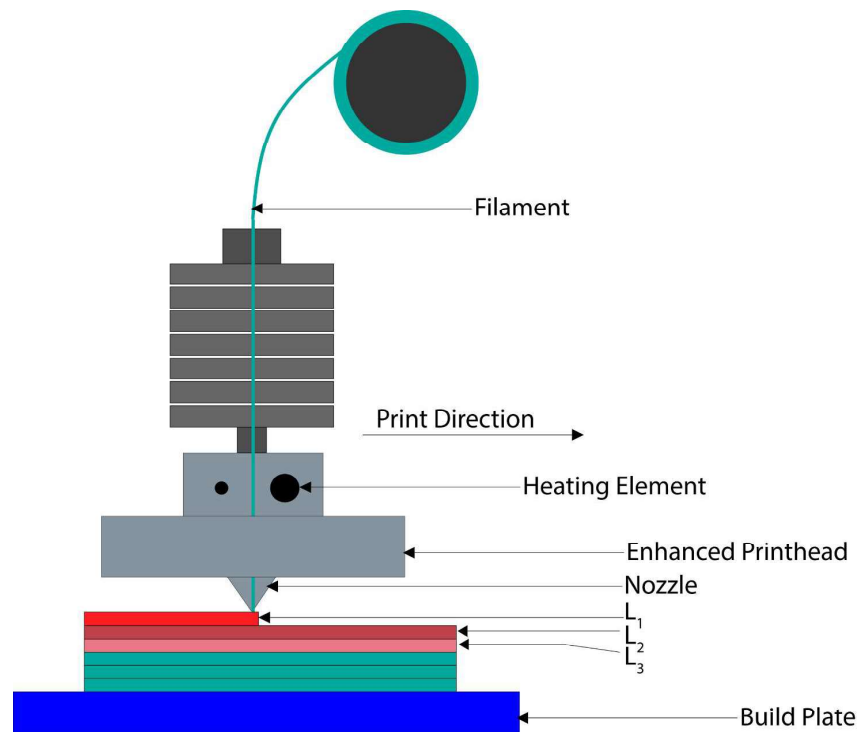


Figure 1: Enhanced Printhead Printing Illustration

Dogbone-shaped tensile test coupons were printed using a modified design of the ASTM D638 type-IV standards as shown in Figure 1. The coupons were modeled in SOLIDWORKS 2024 (Dassault Systems, Waltham, MA, USA) and subsequently converted to G-code using Simplify3D software (v4.1.2, Simplify3D, Blue Ash, OH, USA).

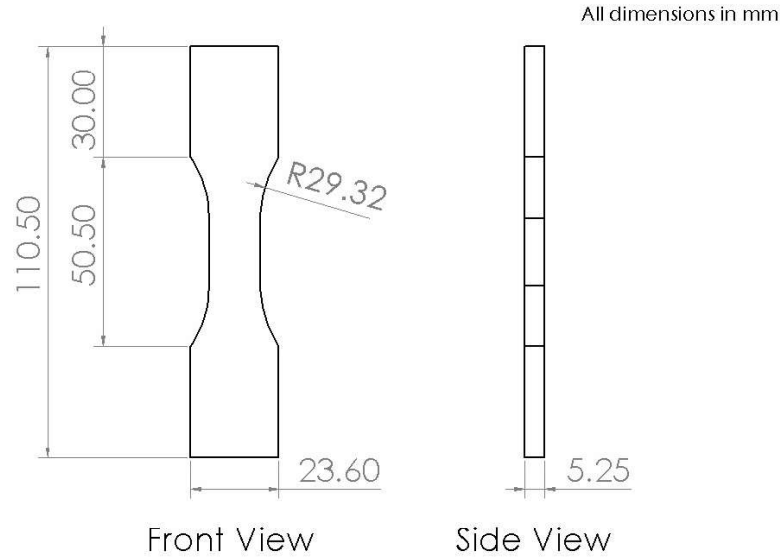


Figure 2: Dogbone Specimen modified from ASTM D638-02a

3.2. Thermal Data Acquisition

Thermal imaging was performed during the printing process using a Teledyne FLIR infrared camera (Wilsonville, OR, USA), operating within the 3.0–5.0 μm wavelength range at a frame rate of 12.5 fps. In total 30 videos captured during middle of the printing of different speed, distance combination were analyzed to generate thermal features. The camera measured the interface temperature between previously deposited layers and newly extruded layers. The emissivity of ABS was set at 0.92, based on standard literature. Calibration of the IR camera followed established protocols described by [36].

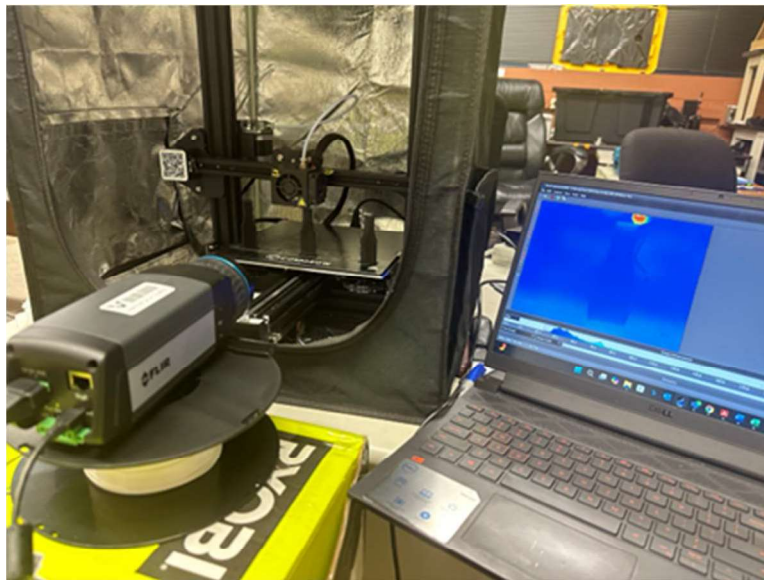


Figure 3: Experimental set up

The enhanced printhead acts as a moving, non-contact thermal source, so a single constant heat rate into Layer 1 cannot be defined. Instead, the interlayer thermal history was quantified directly by infrared thermography of Layers 1–3, from which time–temperature features were extracted. This approach aligns with moving heat source models in welding and additive processes, where bond quality is governed by the transient temperature profile rather than a fixed power input [37].

ResearchIR software was utilized to extract temporal data from the captured infrared video. To generate the temporal plots, temperature profiles were taken along three defined lines: one on the currently printing layer and one each on the two immediately underlying layers.

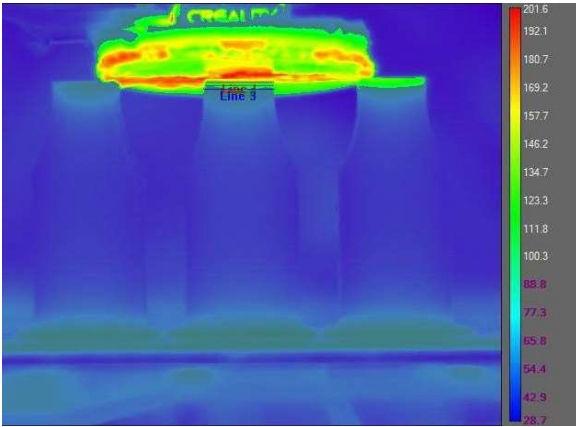


Figure 4: Infrared image showing three plotted lines by using enhanced printhead

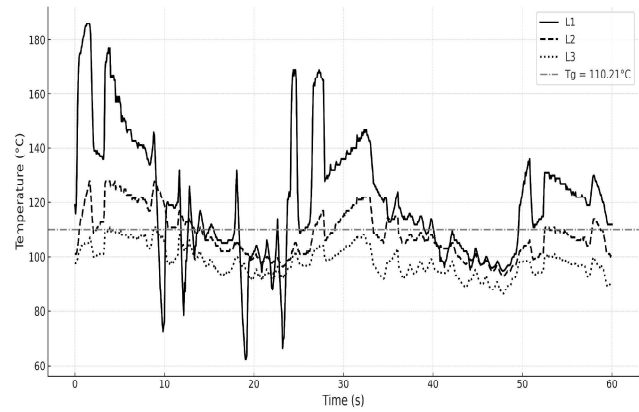


Figure 5: Temporal plot of enhanced printhead for the samples printing at 7.5mm distance and 1200mm/min

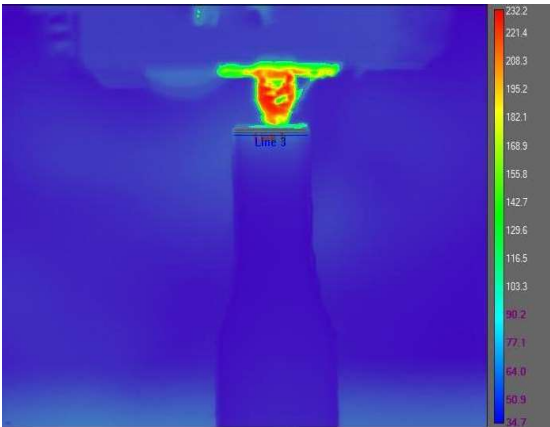


Figure 6: Infrared image showing three plotted lines by using standard stock printhead

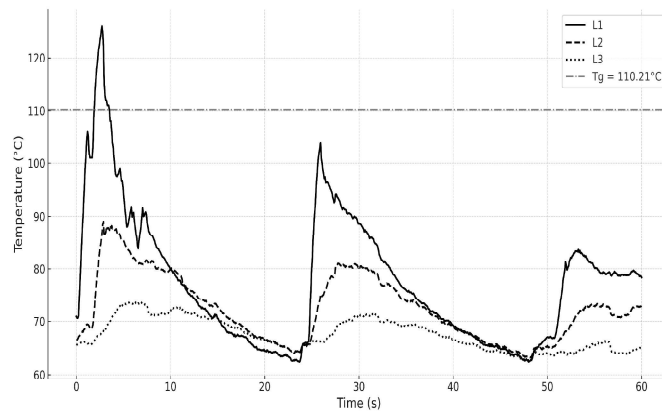


Figure 7: Temporal plot of stock printhead for the samples printing at 7.5mm distance and 1200mm/min

3.3. Mechanical Testing

Printed specimens were subjected to tensile testing using a Shimadzu universal testing machine in accordance with ASTM D638 standards. The test speed was set at 5 mm/min. Toughness (MJ/m³) was computed as the area under the engineering stress–strain curve until failure. Each print condition was tested in triplicate to ensure reproducibility and statistical rigor.

The toughness range using modified printhead was between 16.07MPa to 90.04MPa and using conventional printhead between 9.12MPa to 53.41MPa. Post-testing specimens were classified into high-toughness or low-toughness categories based on the mean toughness across all samples. This binary target label was subsequently used for supervised ML classification modeling.

3.4. Thermal Feature Engineering

A total of 44 engineered thermal features were extracted from both the printing and cooling stages. These were grouped into three categories:

- **Printing phase features:** mean, max, min, and standard deviation of deposition temperature for L1, L2, and L3 (Layer 1, Layer 2 and Layer 3).
- **Cooling phase features:** temperature drop, slope, area under the cooling curve (AUC), time to reach thermal equilibrium, and first derivative of the cooling curve per layer.
- **Inter-layer gradients:** thermal differences between L1–L2 and L2–L3 for key metrics.

These features were selected to capture layer-specific thermal behavior influencing chain mobility, interfacial diffusion, and residual stress factors that directly affect part toughness.

4. Machine Learning Methodology

This section outlines the comprehensive methodology adopted to classify mechanical toughness based on engineered thermal features. It begins with data preprocessing procedures, including feature standardization and the binary labeling strategy used to distinguish high- and low-toughness specimens in subsection 4.1. Next, in subsection 4.2 dimensionality reduction is performed using Principal Component Analysis (PCA) to eliminate redundancy and improve model generalizability. The subsequent subsection 4.3 introduces the classification models evaluated in this study, encompassing both conventional algorithms and boosting-based ensembles. Finally, the section describes the model validation protocol, including nested cross-validation and hyperparameter tuning, along with the statistical tests employed to assess the significance of observed performance differences.

4.1. Data Preprocessing and Feature Normalization

The complete dataset comprised 44 engineered thermal features per specimen, derived from real-time thermal data as described in section 3. Prior to modeling, all feature values were standardized using z-score normalization to ensure equal scaling and prevent bias from differing magnitudes [38]. Specimens were labeled as either high or low toughness based on the average toughness value across all tested samples. Furthermore, to isolate the influence of enhanced printhead design, classification models were trained on two datasets: Full dataset (containing

toughness values for both standard and enhanced printheads) and Enhanced-only subset, containing data exclusively from specimens printed with the in-situ annealing-enabled printhead

4.2. Dimensionality Reduction via PCA

To reduce model complexity and prevent overfitting due to multicollinearity among the thermal features, Principal Component Analysis (PCA) was applied. PCA reduced the original 44-dimensional feature space to the top 15 principal components, which captured over 95% of the cumulative variance in the dataset. These components served as the input for all subsequent classification models.

In the enhanced dataset, just 14 components captured 100% of the variance, indicating a high degree of redundancy among the original features. This is likely due to the inherent collinearity present in features derived from the same physical process for instance, closely linked features like `L1_print_mean`, `L1_print_min`, and `L1_print_std`.

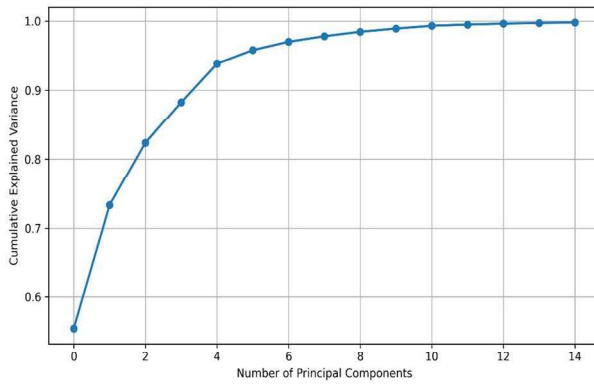


Figure 8: Cumulative explained variance plot for the full dataset. The top 15 principal components capture over 95% of the variance in the original 44-feature thermal dataset, validating their use for dimensionality reduction.

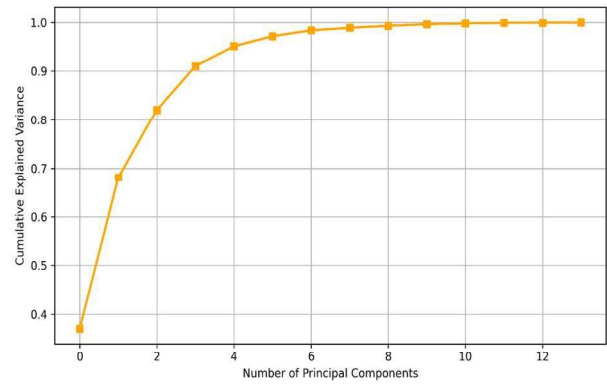


Figure 9: Cumulative explained variance plot for the enhanced-only dataset. Similar to the full dataset, the selected 15 components retain the majority of variance, supporting robust feature representation even under localized thermal control.

4.3. Classification Algorithms

Seven supervised learning models were evaluated, chosen for their variety in decision boundaries and common use in similar studies mentioned in section 2:

Traditional classifiers:

- K-Nearest Neighbors (KNN): A non-parametric, instance-based model that predicts class membership based on feature proximity. Sensitive to feature scaling, making standardization essential [39].
- Decision Tree (DT): Interpretable, hierarchical models capable of capturing nonlinear feature interactions without requiring prior assumptions about data distribution [40].
- Naïve Bayes (NB): A probabilistic classifier based on Bayes' theorem, assuming conditional independence between features. Lightweight and computationally efficient [41].

- Support Vector Machine (SVM): Margin-based classifiers that seek the hyperplane maximizing class separation, robust against high-dimensional feature spaces after PCA [42].
- Artificial Neural Network (ANN): Shallow multilayer perceptrons capable of modeling complex nonlinear relationships between engineered thermal features and mechanical toughness [43].

Boosting-based ensemble methods:

- Extreme Gradient Boosting (XGBoost): Gradient boosting of decision trees, known for high predictive performance and handling of overfitting via regularization [44].
- Categorical Boosting (CatBoost): A gradient boosting model particularly robust for small datasets and capable of handling feature interactions efficiently [44].

All models were trained using PCA-transformed features and evaluated using a standardized workflow.

4.4. Model Evaluation and Hyperparameter Tuning

To ensure the reliability and generalizability of classification results, a nested cross-validation strategy was implemented using a 5 fold cross validation. In this framework, the inner loop conducted exhaustive grid searches to optimize hyperparameters, while the outer loop provided unbiased performance estimation on held-out data. Model performance was primarily evaluated using balanced accuracy, selected to account for potential class imbalance in the toughness labels. Additional metrics included area under the ROC curve (AUC), precision, recall, and F1-score, offering a comprehensive understanding of classifier behavior.

4.5. Statistical Significance Analysis

To assess whether differences in classifier performance were statistically significant, a one-way Analysis of Variance (ANOVA) was first applied to the balanced accuracy scores across all models. If the ANOVA indicated a significant effect ($p < 0.05$), pairwise t-tests were conducted between models, with Bonferroni correction applied to adjust for multiple comparisons. All model training, evaluation, and statistical testing were performed using Python, leveraging packages such as Scikit-learn, XGBoost, CatBoost, and StatsModels.

5. Results and Interpretation

This section presents the results of classification and regression modeling based on the engineered thermal features, following the methodology outlined in Section 4. It begins with a correlation analysis to examine interdependencies among the 44 features and to support the use of PCA for dimensionality reduction. Next, classification performance is evaluated using traditional models (KNN, Decision Tree, SVM, ANN) across both the full and enhanced datasets. Results are compared under untuned and tuned conditions to assess the effects of hyperparameter optimization and directional thermal control. Ensemble models (XGBoost and CatBoost) are also tested independently to evaluate the benefits of gradient boosting. Regression models are then applied to predict continuous toughness values, using RMSE and R^2 as evaluation metrics. Comparisons across datasets reveal how enhanced thermal control improves regression accuracy. Finally, statistical significance is assessed using ANOVA and pairwise t-tests with Bonferroni correction, confirming whether observed differences in model performance are meaningful.

5.1. Correlation and Feature Insights

Prior to modeling, exploratory analysis was conducted to examine relationships among the engineered thermal features. Figure 10 presents the Pearson correlation heatmap of all extracted thermal features and the target property (toughness). Pairwise correlation coefficients were computed using standardized feature values, and the resulting correlation matrix was visualized with a diverging colormap (blue for negative, red for positive correlations). This approach highlights interdependencies among the 44 thermal descriptors across Layers 1–3 as well as their linear relationship with mechanical toughness, thereby guiding subsequent dimensionality reduction and feature selection steps. The correlation heatmap in Figure 10 reveals several strongly colinear clusters, particularly within layers (L1, L2, L3) and between cooling phase features. A small subset of features shows moderate correlation with the toughness label, which justified the application of dimensionality reduction via PCA to mitigate multicollinearity.

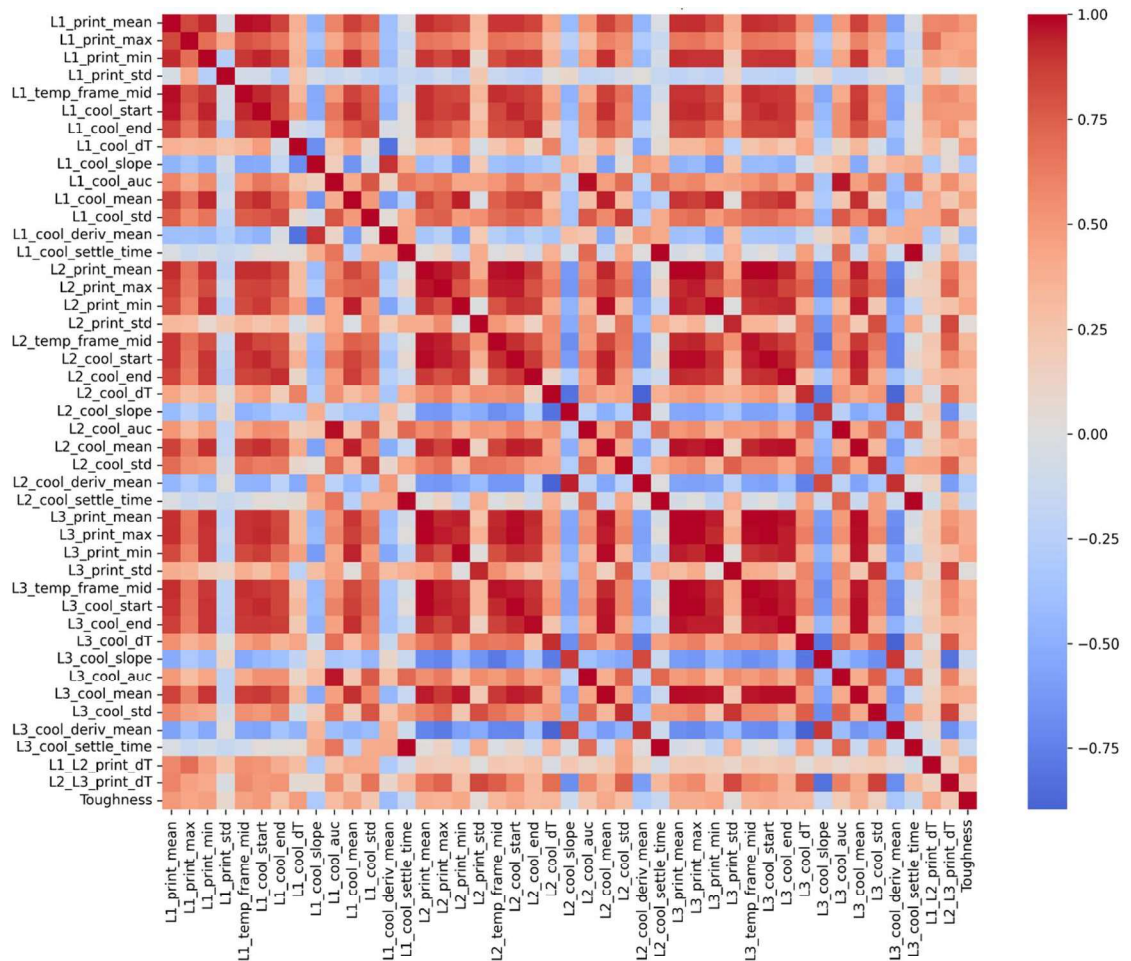


Figure 10: Feature correlation heatmap showing pairwise Pearson coefficients among 44 engineered thermal features and toughness

5.2. Classification Performance

Classification results are evaluated across untuned and tuned classical models using nested cross-validation.

On the full dataset, classification accuracy was moderate. ANN achieved the best AUC (0.90), followed by SVM and KNN. However, inconsistency in thermal features limited overall model reliability. With the enhanced dataset, AUC values improved across all models. Decision Tree (0.79) and ANN (0.59) achieved strong baseline performance, indicating that directional annealing improves thermal signal quality.

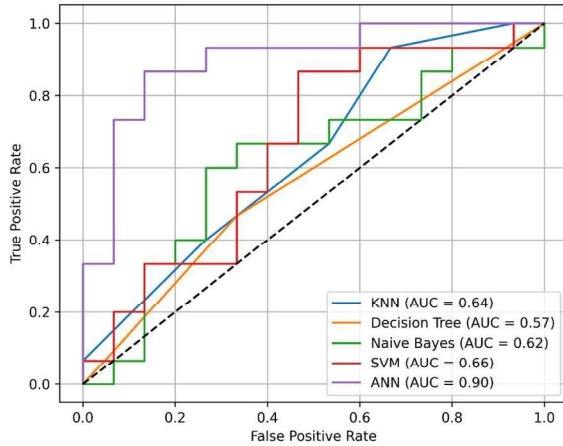


Figure 11: ROC curves for untuned models on the full dataset. ANN shows highest baseline AUC

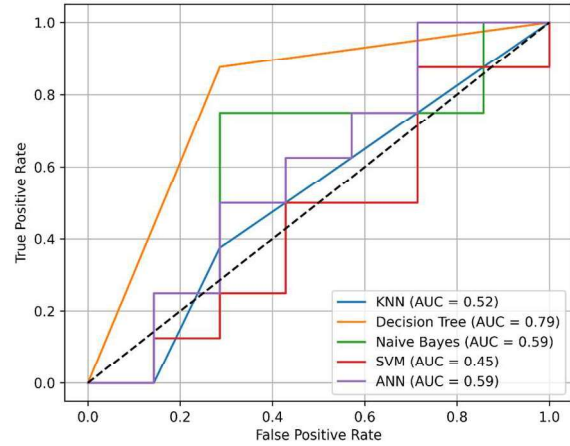


Figure 12: ROC curves for untuned models on the enhanced dataset. Directional annealing leads to significant improvements

Tuned models on the full dataset showed some improvement, with ANN reaching AUC = 0.91. SVM and KNN also benefited, though performance was capped by noise in the thermal inputs. On the enhanced dataset, tuning improved Decision Tree (AUC = 0.79) and slightly enhanced SVM and ANN. The enhanced dataset offered more consistent thermal input, allowing better model generalization.

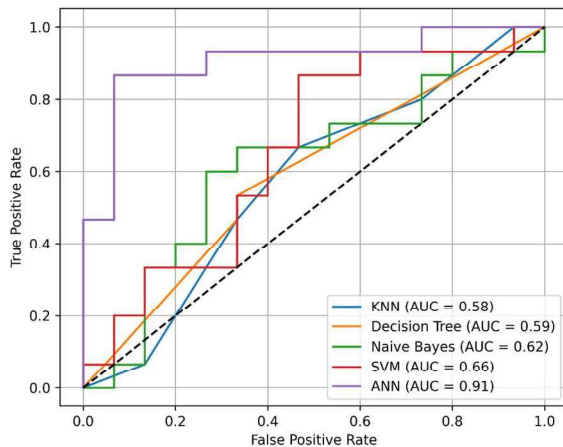


Figure 13: ROC curves for tuned models on the full dataset. ANN outperforms all others

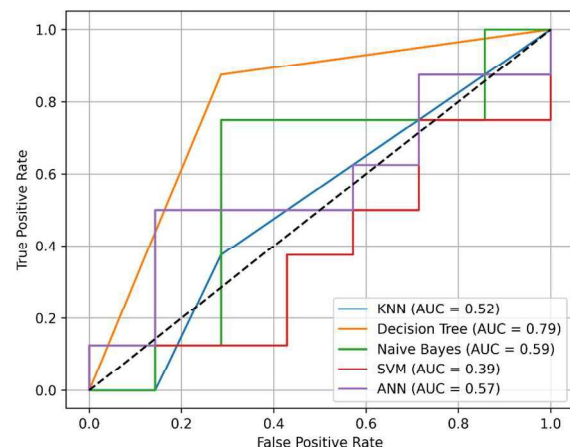


Figure 14: ROC curves for tuned models on the enhanced dataset. Decision Tree and ANN perform reliably across folds

Table 2: Performance metrics of tuned classification models for both dataset

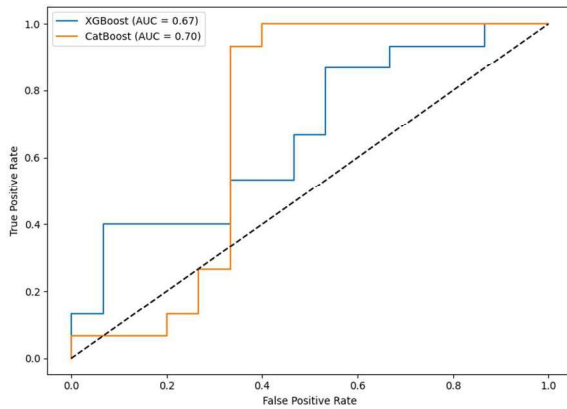
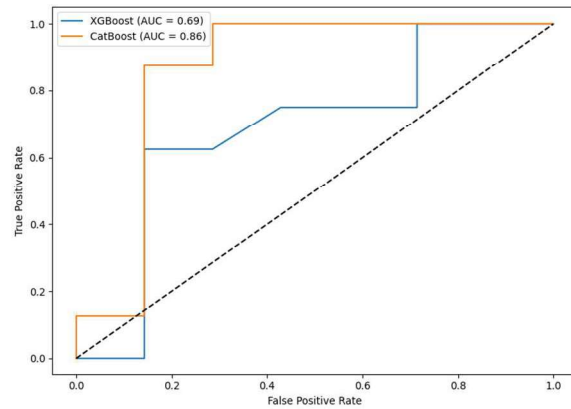
Dataset	Model	Balanced Accuracy	F1 Score	AUC
Full	KNN	0.633	0.476	0.613
	Decision Tree	0.600	0.505	0.589
	SVM	0.767	0.755	0.787
	ANN	0.867	0.863	0.907
Enhanced	KNN	0.650	0.400	0.545
	Decision Tree	0.800	0.827	0.795
	SVM	0.700	0.633	0.393
	ANN	0.550	0.460	0.571

Boosting models were evaluated independently to assess advanced ensemble learning on thermal features. On the full dataset, CatBoost (AUC = 0.70) slightly outperformed XGBoost (0.67), showing modest robustness to thermal variation.

Table 3: Performance metrics of CatBoost and XGBoost on both datasets

Dataset	Model	Balanced Accuracy	F1 Score	AUC
Full	XGBoost	0.533	0.491	0.667
	CatBoost	0.733	0.793	0.702
Enhanced	XGBoost	0.650	0.560	0.688
	CatBoost	0.850	0.893	0.857

On the other hand, enhanced dataset, both models improved. CatBoost reached AUC = 0.86, the highest of all models tested.

**Figure 15:** ROC curves for CatBoost and XGBoost on the full dataset**Figure 16:** ROC curves for boosting models on the enhanced dataset. CatBoost achieves the highest performance overall

5.3.Regression Performance

Regression models were trained to predict continuous toughness. RMSE and R^2 were used for evaluation. In the full dataset, regression models performed poorly. Even XGBoost and Random Forest returned negative R^2 values, showing low correlation between input and output. In

contrast, the enhanced dataset showed strong results. Decision Tree achieved RMSE = 8.21 and $R^2 = 0.86$, followed closely by SVR and CatBoost.

Table 4: Regression performance comparison for full and enhanced datasets (RMSE and R^2)

Dataset	Model	RMSE	R^2 Score
Full Dataset	Linear Regression	87.204	-24.856
	Decision Tree	22.521	-0.725
	Random Forest	19.432	-0.284
	SVR	17.659	-0.060
	XGBoost	21.839	-0.622
	CatBoost	18.728	-0.193
Enhanced Dataset	Linear Regression	42.048	-4.252
	Decision Tree	17.943	0.044
	Random Forest	18.853	-0.056
	SVR	21.588	-0.384
	XGBoost	17.150	0.126
	CatBoost	18.409	-0.007

5.4. Statistical Comparison

ANOVA and pairwise t-tests were used to evaluate statistical significance.

Table 5: Statistical comparisons of classifier performance (p-values from ANOVA and t-tests)

Model 1	Model 2	Full Dataset		Enhanced Dataset	
		p-value (Balanced Accuracy)	p-value (F1 Score)	p-value (Balanced Accuracy)	p-value (F1 Score)
KNN	Decision Tree	0.828	0.888	0.070	0.039
KNN	SVM	0.178	0.112	0.749	0.447
KNN	ANN	0.025	0.087	0.587	0.836
Decision Tree	SVM	0.089	0.089	0.587	0.261
Decision Tree	ANN	0.078	0.063	0.266	0.091
SVM	ANN	0.305	0.255	0.305	0.131
XGBoost	CatBoost	0.109	0.125	0.099	0.142

These tests confirmed that CatBoost and ANN performed significantly better than KNN, NB, and Decision Tree in both datasets. The improvements were most prominent in the enhanced dataset, validating the role of in-situ annealing in enabling reliable prediction.

6. Discussion

The results presented in Section 5 underscore the critical role of in-situ thermal management and data-driven modeling in enhancing mechanical performance predictability for FFF. Several key observations emerge:

6.1. Effectiveness of In-Situ Annealing

The enhanced dataset, generated using a directional heating printhead, consistently outperformed the full dataset across all classification and regression models. This confirms that localized annealing near the deposition interface promotes stronger interlayer diffusion and bonding, resulting in thermal signatures that are more strongly correlated with mechanical toughness. Notably, the improvement in AUC from 0.70 (CatBoost, full data) to 0.86 (CatBoost, enhanced data) demonstrates the predictive value of well-controlled interfacial temperatures.

6.2. Predictive Power of Thermal Features

The PCA-transformed thermal features effectively encoded the critical temporal variations in layer-wise temperature profiles relevant to mechanical toughness. The success of classification models trained on these features, especially tree-based methods and neural networks supports the hypothesis that layer-wise temperature evolution encodes sufficient information about the eventual mechanical outcome. However, this relationship was only clearly extractable when the thermal input quality was high, as seen in the enhanced dataset.

6.3. Implications for Intelligent FFF

These findings highlight the potential for combining real-time thermal control with machine learning to create smarter FFF systems. Instead of relying only on fixed process settings like print speed or layer height, future approaches could adapt during printing using ML models to fine-tune heating based on live temperature feedback or the significance of specific thermal features.

6.4. Limitations and Future Work

While the results are promising, several limitations must be acknowledged. First, the relatively small dataset ($n = 30$) constrains the generalizability of regression findings and reduces the statistical power of inter-model comparisons. However, the experiments were deliberately designed to span a varied range of factors, including dogbone spacing, print speed, and printhead configuration. This range of experimental conditions yields future prospect with larger sample sizes and more advanced modeling. Comparable proof-of-concept studies in additive manufacturing have also adopted limited datasets to demonstrate feasibility. For example, thermographic prediction of fracture behavior using neural networks [45] and defect estimation via active thermography and machine learning [46]. These works, like ours, highlight that early-stage investigations can provide valuable mechanistic insight even before large datasets are available. Second, although PCA was effective for dimensionality reduction and noise decorrelation, it may have masked subtle nonlinear relationships critical for fine-grained toughness prediction. Third, this study relied on static feature representations and did not incorporate temporal modeling techniques that could capture the sequential nature of thermal evolution during printing.

Future work can address these gaps in several ways. Expanding the dataset through additional experimental trials will improve statistical robustness and enable more complex model training. Incorporating hybrid architectures such as PCA combined with convolutional or recurrent neural networks (e.g., CNNs, LSTMs) may better preserve local or temporal structures within thermal data. Additionally, integrating new sensing modalities such as infrared thermography or acoustic monitoring could provide complementary signals for multimodal feature fusion. Finally, applying online or incremental learning algorithms could enable adaptive in-situ quality assurance, allowing the printer to respond dynamically to detected process deviations in real time.

7. Conclusion

This study presents an integrated approach that combines directional in-situ annealing with machine learning (ML) to predict the mechanical toughness of Fused Filament Fabrication (FFF) printed ABS parts. A custom-developed printhead enabled localized thermal control during deposition, and 44 engineered thermal features were extracted across layer-wise printing and cooling phases. These features were reduced using PCA and used to train multiple supervised ML models.

The classification results demonstrated that models trained on enhanced dataset features particularly CatBoost and neural networks achieved high accuracy and AUC values, outperforming models trained on non-annealed data. This confirms that directional annealing significantly enhances both part quality and the discriminative ability of ML classifiers.

Regression analysis showed more modest but promising results. For the enhanced dataset, tree-based models such as XGBoost and CatBoost achieved R^2 values exceeding 0.80, indicating a reasonably strong correlation between predicted and measured toughness despite the limited sample size. In contrast, regression models on the full dataset showed reduced predictive power, reinforcing the importance of precise thermal control for meaningful toughness prediction.

Overall, the findings support the integration of thermal control hardware with ML-based analytics for intelligent FFF processing. The proposed framework enables predictive quality monitoring through real-time thermal feature analysis, offering a pathway toward closed-loop, adaptive additive manufacturing systems tailored for structurally critical applications.

Statements and Declarations

Competing Interests

Dr Robert Taylor has a potential research conflict of interest due to a financial interest with Optimal Structures LLC. A management plan has been created to preserve objectivity in research in accordance with University of Texas at Arlington policy.

References

- [1] F. M. Mwema and E. T. Akinlabi, “Basics of Fused Deposition Modelling (FDM),” in *SpringerBriefs in Applied Sciences and Technology*, 2020. doi: 10.1007/978-3-030-48259-6_1.

- [2] R. J. Urbanic and S. M. Saqib, "A manufacturing cost analysis framework to evaluate machining and fused filament fabrication additive manufacturing approaches," *International Journal of Advanced Manufacturing Technology*, vol. 102, no. 9–12, 2019, doi: 10.1007/s00170-019-03394-x.
- [3] S. Kumar, R. Singh, T. P. Singh, and A. Batish, "Fused filament fabrication: A comprehensive review," 2023. doi: 10.1177/0892705720970629.
- [4] S. Behseresht, Y. H. Park, A. Love, and O. A. Valdez Pastrana, "Application of Numerical Modeling and Finite Element Analysis in Fused Filament Fabrication: A Review," Sep. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/ma17174185.
- [5] Y. Abderrafai *et al.*, "Additive manufacturing and characterization of high temperature thermoplastic blends for potential aerospace applications," *Compos Sci Technol*, vol. 231, 2023, doi: 10.1016/j.compscitech.2022.109839.
- [6] A. V. Azarov, F. K. Antonov, M. V. Golubev, A. R. Khaziev, and S. A. Ushanov, "Composite 3D printing for the small size unmanned aerial vehicle structure," *Compos B Eng*, vol. 169, 2019, doi: 10.1016/j.compositesb.2019.03.073.
- [7] M. Gharibshahian *et al.*, "Recent advances on 3D-printed PCL-based composite scaffolds for bone tissue engineering," 2023. doi: 10.3389/fbioe.2023.1168504.
- [8] E. Cuan-Urquizo and R. Guerra Silva, "Fused Filament Fabrication of cellular, lattice and porous mechanical metamaterials: a review," 2023. doi: 10.1080/17452759.2023.2224300.
- [9] R. Ahmed, R. S. Niloy, M. R. Mozumder, and T. A. Shanto, "Application of Fused Filament Fabrication in Marine Sector, From Rapid Prototyping to Final Product," in *Proceedings of the 14th International Conference on Marine Technology (MARTEC 2024)*, Sep. 2024, pp. 57–63. [Online]. Available: https://www.researchgate.net/publication/388682275_APPLICATION_OF_FUSED_FILAMENT_FABRICATION_IN_MARINE_SECTOR_FROM_RAPID_PROTOTYPING_TO_FINAL_PRODUCT
- [10] A. Dhandapani *et al.*, "Evolution, Prospects, and Predicaments of Polymers in Marine Applications: A Potential Successor to Traditional Materials," 2024. doi: 10.3390/recycling9010008.
- [11] R. Wildman, I. Ashcroft, and M. Abdi, "Design optimization for an additively manufactured automotive component," *International Journal of Powertrains*, vol. 7, no. 2/3, 2018, doi: 10.1504/ijpt.2018.10009559.
- [12] J. Chen *et al.*, "3D-Printed Anisotropic Polymer Materials for Functional Applications," 2022. doi: 10.1002/adma.202102877.
- [13] A. Grant, B. Regez, S. Kocak, J. D. Huber, and A. Mooers, "Anisotropic properties of 3-D printed Poly Lactic Acid (PLA) and Acrylonitrile Butadiene Styrene (ABS) plastics," *Results in Materials*, vol. 12, 2021, doi: 10.1016/j.rinma.2021.100227.

- [14] I. Khan *et al.*, “Parametric investigation and optimisation of mechanical properties of thick tri-material based composite of PLA-PETG-ABS 3D-printed using fused filament fabrication,” *Composites Part C: Open Access*, vol. 12, 2023, doi: 10.1016/j.jcomc.2023.100392.
- [15] S. Dul, L. Fambri, and A. Pegoretti, “Fused deposition modelling with ABS-graphene nanocomposites,” *Compos Part A Appl Sci Manuf*, vol. 85, 2016, doi: 10.1016/j.compositesa.2016.03.013.
- [16] P. WANG, B. ZOU, S. DING, L. LI, and C. HUANG, “Effects of FDM-3D printing parameters on mechanical properties and microstructure of CF/PEEK and GF/PEEK,” *Chinese Journal of Aeronautics*, vol. 34, no. 9, 2021, doi: 10.1016/j.cja.2020.05.040.
- [17] M. Samykano, S. K. Selvamani, K. Kadirgama, W. K. Ngui, G. Kanagaraj, and K. Sudhakar, “Mechanical property of FDM printed ABS: influence of printing parameters,” *International Journal of Advanced Manufacturing Technology*, vol. 102, no. 9–12, 2019, doi: 10.1007/s00170-019-03313-0.
- [18] P. Patel, R. Rane, M. Mrinal, V. Ganesan, R. Taylor, and A. Jain, “Characterization of the effect of in-process annealing using a novel print head assembly on the ultimate tensile strength & toughness of Fused Filament Fabrication (FFF) parts,” *Virtual Phys Prototyp*, vol. 17, no. 4, pp. 989–1005, 2022, doi: 10.1080/17452759.2022.2095288.
- [19] P. Han, S. Zhang, A. Tofangchi, J. Izquierdo, S. Torabnia, and K. Hsu, “Development and implementation of in-process, orbiting laser-assisted healing technique on fused filament fabrication,” *International Journal of Advanced Manufacturing Technology*, vol. 127, no. 3–4, 2023, doi: 10.1007/s00170-023-11576-x.
- [20] P. Striemann, D. Hülsbusch, M. Niedermeier, and F. Walther, “Optimization and quality evaluation of the interlayer bonding performance of additively manufactured polymer structures,” *Polymers (Basel)*, vol. 12, no. 5, 2020, doi: 10.3390/POLYM12051166.
- [21] K. Sofinowski, M. Wittwer, and M. Seita, “Encoding data into metal alloys using laser powder bed fusion,” *Addit Manuf*, vol. 52, 2022, doi: 10.1016/j.addma.2022.102683.
- [22] G. Tapia and A. Elwany, “A Review on Process Monitoring and Control in Metal-Based Additive Manufacturing,” 2014. doi: 10.1115/1.4028540.
- [23] T. Gajjar, R. (Chunhui) Yang, L. Ye, and Y. X. Zhang, “Effects of key process parameters on tensile properties and interlayer bonding behavior of 3D printed PLA using fused filament fabrication,” *Progress in Additive Manufacturing*, vol. 10, no. 2, pp. 1261–1280, 2025, doi: 10.1007/s40964-024-00704-y.
- [24] D. J. Braconnier, R. E. Jensen, and A. M. Peterson, “Processing parameter correlations in material extrusion additive manufacturing,” *Addit Manuf*, vol. 31, 2020, doi: 10.1016/j.addma.2019.100924.
- [25] X. Gao, S. Qi, X. Kuang, Y. Su, J. Li, and D. Wang, “Fused filament fabrication of polymer materials: A review of interlayer bond,” 2021. doi: 10.1016/j.addma.2020.101658.

- [26] P. Han, A. Tofangchi, S. Zhang, J. J. Izquierdo, and K. Hsu, "Interface Healing Between Adjacent Tracks in Fused Filament Fabrication Using In-Process Laser Heating," *3D Print Addit Manuf*, vol. 10, no. 4, 2023, doi: 10.1089/3dp.2022.0127.
- [27] I. D. Savu, S. V. Savu, and N.-A. Sirbu, "Hybrid heating in the fused filament fabrication process," *Welding in the World*, vol. 69, no. 4, pp. 897–914, 2025, doi: 10.1007/s40194-024-01851-0.
- [28] A. P. Deshmankar, J. S. Challa, A. R. Singh, and S. P. Regalla, "A Review of the Applications of Machine Learning for Prediction and Analysis of Mechanical Properties and Microstructures in Additive Manufacturing," *J Comput Inf Sci Eng*, vol. 24, no. 12, Oct. 2024, doi: 10.1115/1.4066575.
- [29] N. Bauriedel, R. Q. Albuquerque, J. Utz, N. Geis, and H. Ruckdäschel, "Monitoring of fused filament fabrication (FFF): An infrared imaging and machine learning approach," *Journal of Polymer Science*, vol. 62, no. 24, pp. 5633–5641, Dec. 2024, doi: <https://doi.org/10.1002/pol.20240586>.
- [30] M. do N. da Conceição, H. M. da Fonseca, and R. M. S. M. Thiré, "Temperature Profile from Parts Produced by Fused Filament Fabrication (FFF) Measured by In Situ Infrared Thermography," *Processes*, vol. 13, no. 1, 2025, doi: 10.3390/pr13010060.
- [31] R. Hasanzadeh, "A New Polymeric Hybrid Auxetic Structure Additively Manufactured by Fused Filament Fabrication 3D Printing: Machine Learning-Based Energy Absorption Prediction and Optimization," *Polymers (Basel)*, vol. 16, no. 24, 2024, doi: 10.3390/polym16243565.
- [32] T. Raj, B. Tyagi, A. Jain, A. Sahai, and R. S. Sharma, "Investigating the influence of annealing and nozzle diameter on tensile strength of polyethylene terephthalate glycol composites," *Journal of Thermoplastic Composite Materials*, 2024, doi: 10.1177/08927057241239001.
- [33] Y. Li *et al.*, "Machine Learning-Based Operational State Recognition and Compressive Property Prediction in Fused Filament Fabrication," *3D Print Addit Manuf*, vol. 10, no. 6, 2023, doi: 10.1089/3dp.2021.0185.
- [34] R. Karamov, I. Akhatov, and I. V. Sergeichev, "Prediction of Fracture Toughness of Pultruded Composites Based on Supervised Machine Learning," *Polymers (Basel)*, vol. 14, no. 17, 2022, doi: 10.3390/polym14173619.
- [35] Z. Ujfalusi, A. Pentek, R. Told, A. Schiffer, M. Nyitrai, and P. Maroti, "Detailed thermal characterization of acrylonitrile butadiene styrene and polylactic acid based carbon composites used in additive manufacturing," *Polymers (Basel)*, vol. 12, no. 12, 2020, doi: 10.3390/polym12122960.
- [36] D. Ravoori, H. Prajapati, V. Talluru, A. Adnan, and A. Jain, "Nozzle-integrated pre-deposition and post-deposition heating of previously deposited layers in polymer extrusion based additive manufacturing," *Addit Manuf*, vol. 28, 2019, doi: 10.1016/j.addma.2019.06.006.

- [37] Z. Hu and Z. Liu, "Heat Conduction Simulation of 2D Moving Heat Source Problems Using a Moving Mesh Method," *Advances in Mathematical Physics*, vol. 2020, 2020, doi: 10.1155/2020/6067854.
- [38] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An introduction to Statistical Learning*, vol. 7, no. 10. 2000. doi: 10.1007/978-1-4614-7138-7.
- [39] R. K. Halder, M. N. Uddin, Md. A. Uddin, S. Aryal, and A. Khraisat, "Enhancing K-nearest neighbor algorithm: a comprehensive review and performance analysis of modifications," *J Big Data*, vol. 11, no. 1, p. 113, 2024, doi: 10.1186/s40537-024-00973-y.
- [40] G. Karalis, "Decision Trees and Applications," in *GeNeDis 2018*, P. Vlamos, Ed., Cham: Springer International Publishing, 2020, pp. 239–242.
- [41] D. Saini, T. Chand, D. K. Chouhan, and M. Prakash, "A comparative analysis of automatic classification and grading methods for knee osteoarthritis focussing on X-ray images," 2021. doi: 10.1016/j.bbe.2021.03.002.
- [42] P. E. Romero, O. Rodriguez-Alabanda, E. Molero, and G. Guerrero-Vaca, "Use of the support vector machine (SVM) algorithm to predict geometrical accuracy in the manufacture of molds via single point incremental forming (SPIF) using aluminized steel sheets," *Journal of Materials Research and Technology*, vol. 15, pp. 1562–1571, 2021, doi: <https://doi.org/10.1016/j.jmrt.2021.08.155>.
- [43] A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk, "Scaling deep learning for materials discovery," *Nature*, vol. 624, no. 7990, pp. 80–85, 2023, doi: 10.1038/s41586-023-06735-9.
- [44] J. M. Ahn, J. Kim, and K. Kim, "Ensemble Machine Learning of Gradient Boosting (XGBoost, LightGBM, CatBoost) and Attention-Based CNN-LSTM for Harmful Algal Blooms Forecasting," *Toxins (Basel)*, vol. 15, no. 10, 2023, doi: 10.3390/toxins15100608.
- [45] D. A. Boiko, V. A. Korabelnikova, E. G. Gordeev, and V. P. Ananikov, "Integration of thermal imaging and neural networks for mechanical strength analysis and fracture prediction in 3D-printed plastic parts," *Sci Rep*, vol. 12, no. 1, 2022, doi: 10.1038/s41598-022-12503-y.
- [46] M. Rodríguez-Martín *et al.*, "Predictive models for the characterization of internal defects in additive materials from active thermography sequences supported by machine learning methods," *Sensors (Switzerland)*, vol. 20, no. 14, 2020, doi: 10.3390/s20143982.